■ GREENAIR ENVIRONMENTAL INTELLIGENCE

GreenAir Executive Environmental Assessment

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1. Executive Summary & Findings

• Analyzed dataset 'synthetic_air_quality_demo.csv' with 14 total records.

• Data quality health index evaluated at 100.0/100.

• PM2.5 time-series trend is 'Increasing' with net change of 7.39%.

• Identified 1 statistical anomaly readings in PM2.5 concentrations.

3. Dataset Quality & Descriptive Statistics

Filename:	synthetic_air_quality_demo.csv
Observations:	14 records (9 columns)
Data Quality Score:	100.0/100
Date Span:	2026-10-01T08:00:00 to 2026-10-14T08:00:00

4. Time-Series Trend Analysis

Pollutant: **PM2.5** | Direction: **Increasing** | Rate of Change: 7.39% Analyzed 14 chronological observations from 2026-10-01 08:00:00 to 2026-10-14 08:00:00.

5. Anomaly Detection Summary

Examined 14 records using IQR method. Identified **1** anomalous event(s) (7.14% of dataset). Acceptable range: [6.61, 80.31].

Regulatory Notice & Limitations

This report is generated autonomously by the GreenAir AI Agent for analytical, educational, and decision-support purposes. Model classifications reflect patterns learned by a supervised machine-learning model from training data and do not substitute for official regulatory Air Quality Index (AQI) determinations published by government environmental authorities. Statistical anomalies indicate mathematical outliers in sample distributions and should not be interpreted as confirmed sensor failures or emergency incidents without ground-truth physical verification. This document does not provide personalized medical diagnoses.